

Module 3 : Transportation Problem : General structure of transportation problem, methods of finding initial basic feasible solution (NWCM, LCM & VAM), Degeneracy, Optimality Test using Stepping Stone and MODI Methods (theory only).

TRANSPORTATION PROBLEM

I. **Meaning** : The study of **transportation problem** helps to identify optimal transportation routes along with units of commodity to be shipped in order to minimize total transportation cost.

II. **Introduction** : The structure of transportation problem involves a large number of shipping routes from several supply centres to several demand centres. Thus, objective is to determine shipping routes between supply centres and demand centres in order to satisfy the required quantity of goods or services at each destination centre, with available quantity of goods or services at each supply centre at the minimum transportation cost and/ or time.

The transportation algorithms help to minimize the total cost of transporting a homogeneous commodity (product) from supply centres to demand centres. However, it can also be applied to the maximization of total value or utility.

III. GENERAL MATHEMATICAL MODEL OF TRANSPORTATION PROBLEM

Let there be m sources of supply, $S_1, S_2, \dots, S_m$ having a_i ($i = 1, 2, \dots, m$) units of supply (or capacity), respectively to be transported to n destinations, $D_1, D_2, \dots, D_n$ with b_j ($j = 1, 2, \dots, n$) units of demand (or requirement), respectively. Let c_{ij} be the cost of shipping one unit of the commodity from source i to destination j . If x_{ij} represents number of units shipped from source i to destination j , the problem is to determine the transportation schedule so as to minimize the total transportation cost while satisfying the supply and demand conditions. Mathematically, the transportation problem, in general, may be stated as follows:

$$\text{Minimize (total cost) } Z = \sum_{i=1}^m \sum_{j=1}^n c_{ij}x_{ij} \quad (1)$$

subject to the constraints

$$\sum_{j=1}^n x_{ij} = a_i, i = 1, 2, \dots, m \text{ (supply constraints)} \quad (2)$$

$$\sum_{i=1}^m x_{ij} = b_j, j = 1, 2, \dots, n \text{ (demand constraints)} \quad (3)$$

and $x_{ij} \geq 0$ for all i and j .

IV. POINTS TO REMEMBER

- **Balanced and Unbalanced Problem** : When the total supply is equal to the total demand, the problem is called a **balanced transportation problem**, otherwise it is called an **unbalanced transportation problem**. The unbalanced transportation problem can be made balanced by adding a dummy supply centre (row) or a dummy demand centre (column) as the need arises.
- When the number of positive allocations (values of decision variables) at any stage of the feasible solution is less than the required number (rows + columns – 1), i.e. number of independent constraint equations, the solution is said to be **degenerate**, otherwise **non-degenerate**.
- Cells in the transportation table having positive allocation, i.e., $x_{ij} > 0$ are called **occupied cells**, otherwise are known as **non-occupied (or empty) cells**.

V. THE TRANSPORTATION ALGORITHM

The algorithm for solving a transportation problem may be summarized into the following steps:

Step 1: Formulate the problem and arrange the data in the matrix form

The formulation of the transportation problem is similar to the LP problem formulation. In transportation problem, the objective function is the total transportation cost and the constraints are the amount of supply and demand available at each source and destination, respectively.

Step 2: Obtain an initial basic feasible solution

Following three different methods are discussed to obtain an initial solution:

- North-West Corner Method,
- Least Cost Method, and
- Vogel's Approximation (or Penalty) Method.

The initial solution obtained by any of the three methods must satisfy the following conditions:

- (i) The solution must be feasible, i.e. it must satisfy all the supply and demand constraints (also called *rim conditions*).
- (ii) The number of positive allocations must be equal to $m + n - 1$, where m is the number of rows and n is the number of columns. Any solution that satisfies the above conditions is called *non-degenerate basic feasible solution*, otherwise, *degenerate solution*.

Step 3: Test the initial solution for optimality

The Modified Distribution (MODI) method is discussed to test the optimality of the solution obtained in Step 2. If the current solution is optimal, then stop. Otherwise, determine a new improved solution.

Step 4: Updating the solution : Repeat Step 3 until an optimal solution is reached.

VI. METHODS OF FINDING INITIAL SOLUTION

There are several methods available to obtain an initial basic feasible solution. But according to syllabus we shall discuss only following three methods:

- A. North-West Corner Method,
- B. Least Cost Method, and
- C. Vogel's Approximation (or Penalty) Method.

A. NORTH-WEST CORNER METHOD (NWCM)

This method does not take into account the cost of transportation on any route of transportation. The method can be summarized as follows:

Step 1:

Start with the cell at the upper left (north-west) corner of the transportation table (or matrix) and allocate commodity equal to the minimum of the rim values for the first row and first column, i.e. $\min(a_1, b_1)$.

Step 2:

- (a) If allocation made in Step 1 is equal to the supply available at first source (a_1 , in first row), then move vertically down to the cell (2, 1), i.e., second row and first column. Apply Step 1 again, for next allocation.
- (b) If allocation made in Step 1 is equal to the demand of the first destination (b_1 in first column), then move horizontally to the cell (1, 2), i.e., first row and second column. Apply Step 1 again for next allocation.
- (c) If $a_1 = b_1$, allocate $x_{11} = a_1$ or b_1 and move diagonally to the cell (2, 2).

Step 3: Continue the procedure step by step till an allocation is made in the south-east corner cell of the transportation table.

Remark : If during the process of making allocation at a particular cell, the supply equals demand, then the next allocation of magnitude zero can be made in a cell either in the next row or column. This condition is known as degeneracy.

B. LEAST COST METHOD (LCM)

Since the main objective is to minimize the total transportation cost, transport as much as possible through those routes (cells) where the unit transportation cost is lowest. This method takes into account the minimum unit cost of transportation for obtaining the initial solution and can be summarized as follows:

Step 1: Select the cell with the lowest unit cost in the entire transportation table and allocate as much as possible to this cell. Then eliminate (line out) that row or column in which either the supply or demand is fulfilled. If a row and a column are both satisfied simultaneously, then crossed off either a row or a column. In case the smallest unit cost cell is not unique, then select the cell where the maximum allocation can be made.

Step 2: After adjusting the supply and demand for all uncrossed rows and columns repeat the procedure to select a cell with the next lowest unit cost among the remaining rows and columns of the transportation table and allocate as much as possible to this cell. Then crossed off that row and column in which either supply or demand is exhausted.

Step 3: Repeat the procedure until the available supply at various sources and demand at various destinations is satisfied. The solution so obtained need not be non-degenerate.

C. Vogel's Approximation Method (VAM)

Vogel's approximation (penalty or regret) is preferred over NWCR and LCM methods. In this method, an allocation is made on the basis of the opportunity (or penalty or extra) cost that would have been incurred if the allocation in certain cells with minimum unit transportation cost were missed. Hence, allocations are made in such a way that the penalty cost is minimized. An initial solution obtained by using this method is nearer to an optimal solution or is the optimal solution itself. The steps of VAM are as follows:

Step 1: Calculate the penalties for each row (column) by taking the difference between the smallest and next smallest unit transportation cost in the same row (column). This difference indicates the penalty or extra cost that has to be paid if decision-maker fails to allocate to the cell with the minimum unit transportation cost.

Step 2: Select the row or column with the largest penalty and allocate as much as possible in the cell that has the least cost in the selected row or column and satisfies the rim conditions. If there is a tie in the values of penalties, it can be broken by selecting the cell where the maximum allocation can be made.

Step 3: Adjust the supply and demand and cross out the satisfied row or column. If a row and a column are satisfied simultaneously, only one of them is crossed out and the remaining row (column) is assigned a zero supply (demand). Any row or column with zero supply or demand should not be used in computing future penalties.

Step 4: Repeat Steps 1 to 3 until the available supply at various sources and demand at various destinations is satisfied.

VII. TEST FOR OPTIMALITY

Once an initial solution is obtained, the next step is to check its optimality in terms of feasibility of the solution and total minimum transportation cost.

The test of optimality begins by calculating an opportunity cost associated with each unoccupied cell (represents unused route) in the transportation table. An unoccupied cell with the largest negative opportunity cost is selected to include in the new set of transportation routes (allocations).

This value indicates the per unit cost reduction that can be achieved by making appropriate allocation in the unoccupied cell. This cell is also known as an *incoming cell (or variable)*. The outgoing cell (or variable) from the current solution is the occupied cell (basic variable) where allocation will become zero as allocation is made in the unoccupied cell with the largest negative opportunity cost. Such an exchange reduces the total transportation cost. The process is continued until there is no negative opportunity cost. That is, the current solution is an optimal solution.

The **Modified-distribution (MODI)** method (also called *u-v method* or *method of multipliers*) is used to calculate opportunity cost associated with each unoccupied cell and then improving the current solution leading to an optimal solution.

STEPS OF MODI METHOD (TRANSPORTATION ALGORITHM)

Step 1: For an initial basic feasible solution with $m + n - 1$ occupied cells, calculate u_i and v_j for rows and columns. The initial solution can be obtained by any of the three methods discussed earlier. To start with, any one of u_i s or v_j s is assigned the value zero. It is better to assign zero to a particular u_i or v_j where there are maximum number of allocations in a row or column respectively, as this will reduce the considerably arithmetic work. The value of u_i s and v_j s for other rows and columns is calculated by using the relationship. $c_{ij} = u_i + v_j$, for all occupied cells (i, j) .

Step 2: For unoccupied cells, calculate the opportunity cost by using the relationship $d_{ij} = c_{ij} - (u_i + v_j)$, for all i and j .

Step 3: Examine sign of each d_{ij}

- (i) If $d_{ij} > 0$, then the current basic feasible solution is optimal.
- (ii) If $d_{ij} = 0$, then the current basic feasible solution will remain unaffected but an alternative solution exists.
- (iii) If one or more $d_{ij} < 0$, then an improved solution can be obtained by entering an unoccupied cell (i, j) into the solution mix (basis). An unoccupied cell having the largest negative value of d_{ij} is chosen for entering into the solution mix (new transportation schedule).

Step 4: Construct a closed-path (or loop) for the unoccupied cell with largest negative value of d_{ij} . Start the closed path with the selected unoccupied cell and mark a plus sign (+) in this cell. Trace a path along the rows (or columns) to an occupied cell, mark the corner with a minus sign (–) and continue down the column (or row) to an occupied cell. Then mark the corner with plus sign (+) and minus sign (–) alternatively. Close the path back to the selected unoccupied cell.

Step 5: Select the smallest quantity amongst the cells marked with minus sign on the corners of closed loop. Allocate this value to the selected unoccupied cell, add it to occupied cells marked with plus signs, and subtract it from the occupied cells marked with minus signs.

Step 6: Obtain a new improved solution by allocating units to the unoccupied cell according to Step 5 and calculate the new total transportation cost.

Step 7: Test optimality of the revised solution. The procedure terminates when all $d_{ij} \geq 0$ for unoccupied cell